

[TGRS 2025] DRPCA-Net: Make Robust PCA Great Again for Infrared Small Target Detection

Zihao Xiong, Fei Zhou, Fengyi Wu, Shuai Yuan,
Maixia Fu, Zhenming Peng, Jian Yang, Yimian Dai



Arxiv: <https://arxiv.org/abs/2507.09541>

Code: <https://github.com/GrokCV/DRPCA-Net>

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1. 背景

红外小目标检测意义:

- 军事防御与预警
- 民用与安全



模型驱动:

优:

- 理论与物理基础
- 可解释性

缺:

- 泛化性较差

平衡

深度展开网络
DUNs

数据驱动:

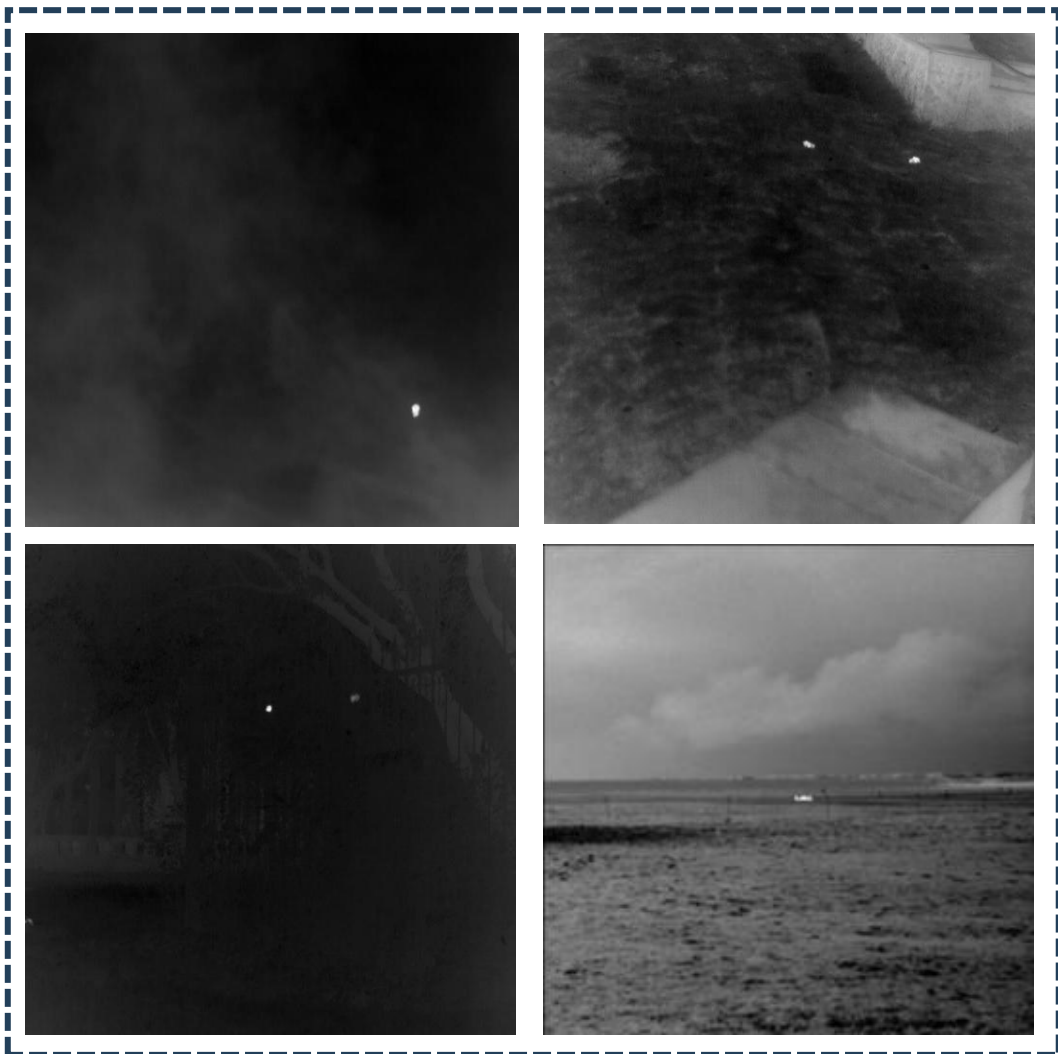
优:

- 强大的特征提取与表示能力
- 高检测精度

缺:

- “黑盒”

2. 动机

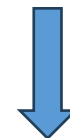


红外小目标:

- 占用小于 81 像素或小于 0.15% 面积
- 缺乏形状和纹理
- 信噪比低



★ 目标具有强稀疏性



- 目标稀疏先验刻画不足

深度展开网络
DUNs

2. 动机

稳健主成分分析 (RPCA) :

$$M = L + S$$

低秩 稀疏

优化问题:

$$\text{minimize } \|L\|_* + \lambda \|S\|_1 \text{ s.t. } L + S = M$$

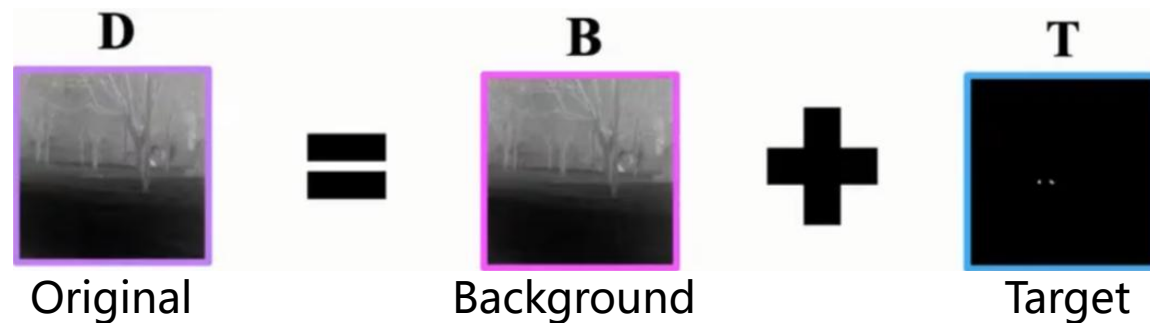


有约束: $\min_{B, T} \|B\|_* + \lambda \|T\|_1 \text{ s.t. } D = B + T$

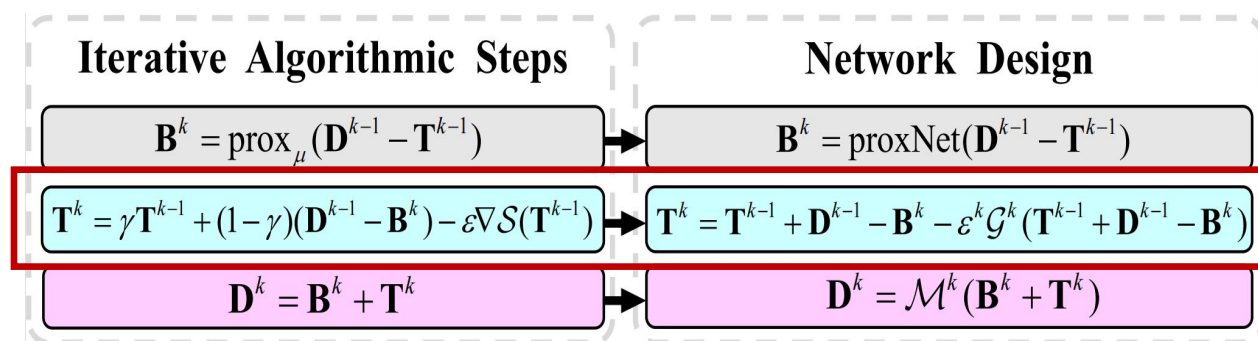


无约束: $\mathcal{L}(B, T) = \mathcal{R}(B) + \lambda \mathcal{S}(T) + \frac{\mu}{2} \|D - B - T\|_F^2$

RPCA Modeling:

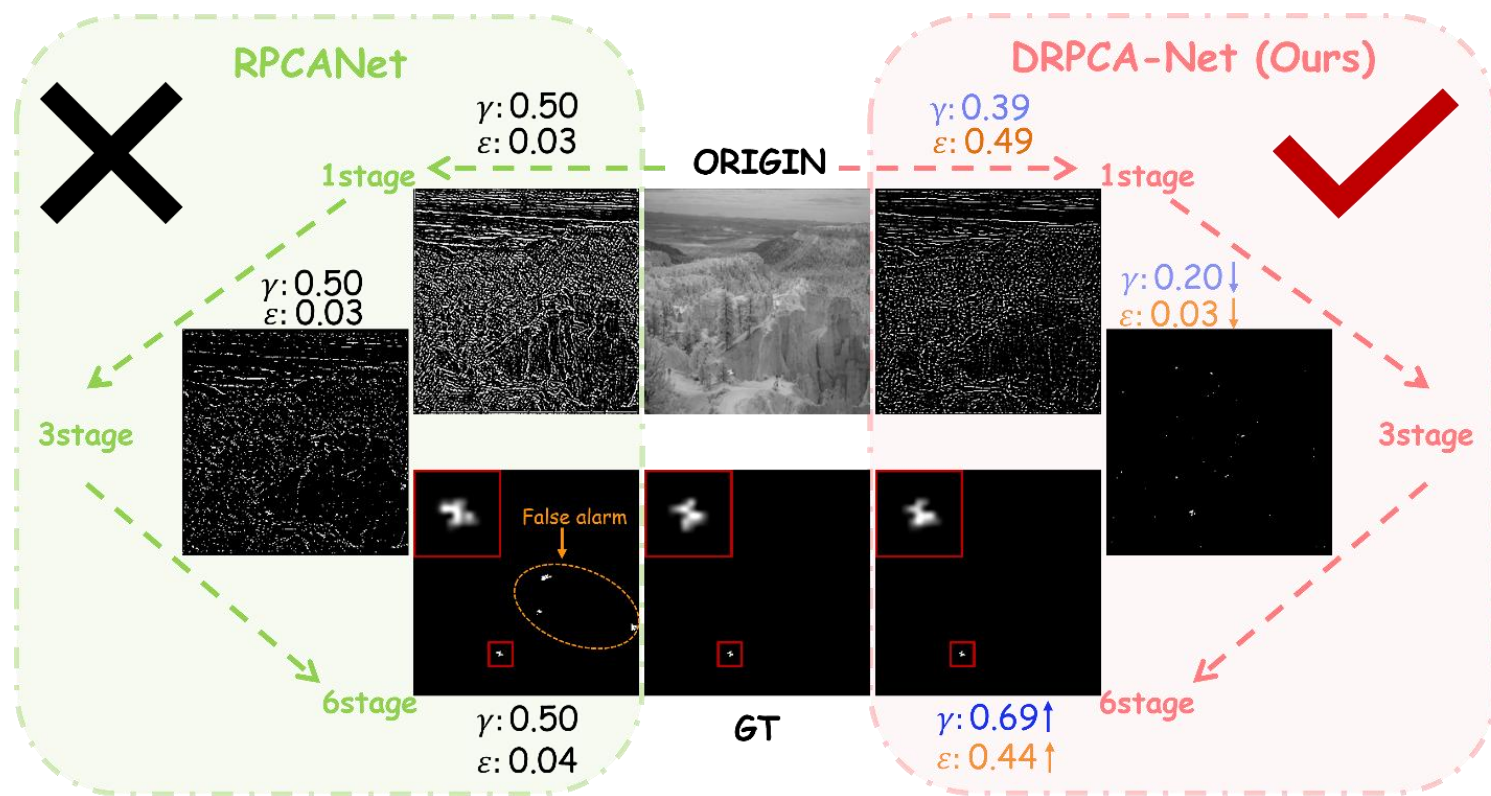


RPCANet [2] :



[2] F. Wu, T. Zhang, L. Li, Y. Huang, and Z. Peng, "RPCANet: Deep unfolding RPCA based infrared small target detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 4797–4806.

2. 动机



动机:

- 目标稀疏先验刻画不足
- 无场景感知的静态超参数

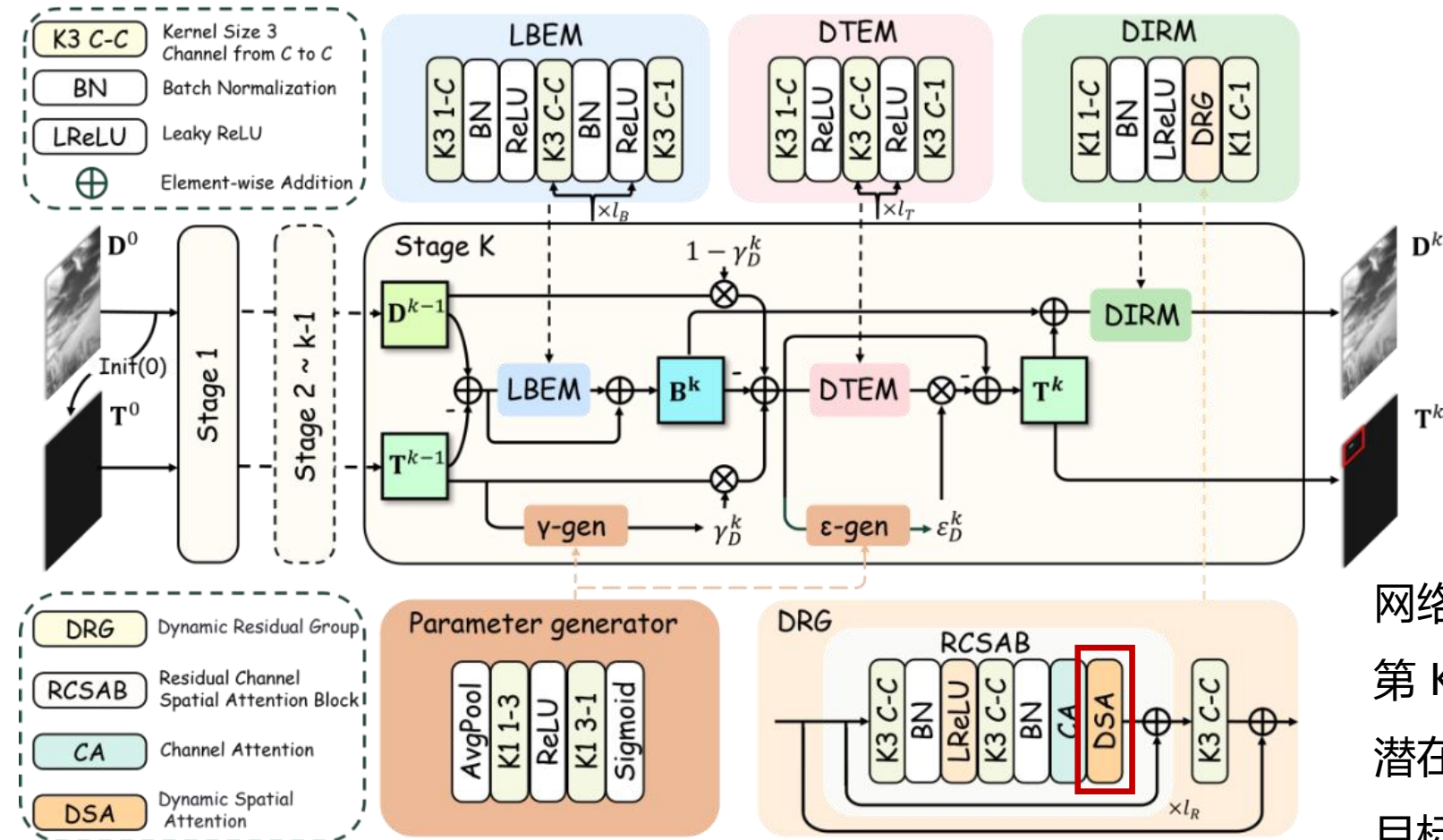
贡献:

- 场景感知的动态超参数
- 动态残差组 (DRG)

传统深度展开和我们提出的动态展开的不同阶段（1、3 和 6）可视化表明，动态参数生成机制有助于有效抑制背景。

3. DRPCA-Net

DRPCA-Net 的整体结构图:



DRPCA-Net 迭代公式:

$$\mathbf{B}^k = (\mathbf{D}^{k-1} - \mathbf{T}^{k-1}) + \mathcal{G}_{\theta}^k(\mathbf{D}^{k-1} - \mathbf{T}^{k-1})$$

$$\mathbf{T}^k = \gamma_D^k \mathbf{T}^{k-1} + (1 - \gamma_D^k)(\mathbf{D}^{k-1} - \mathbf{B}^k) - \varepsilon_D^k \mathcal{S}^k(\gamma_D^k \mathbf{T}^{k-1} + (1 - \gamma_D^k)(\mathbf{D}^{k-1} - \mathbf{B}^k))$$

$$\mathbf{D}^k = \mathcal{DM}^k(\mathbf{B}^k + \mathbf{T}^k)$$

网络由 K 阶段组成, 每个阶段的结构是相同的
第 K 阶段的详细结构:

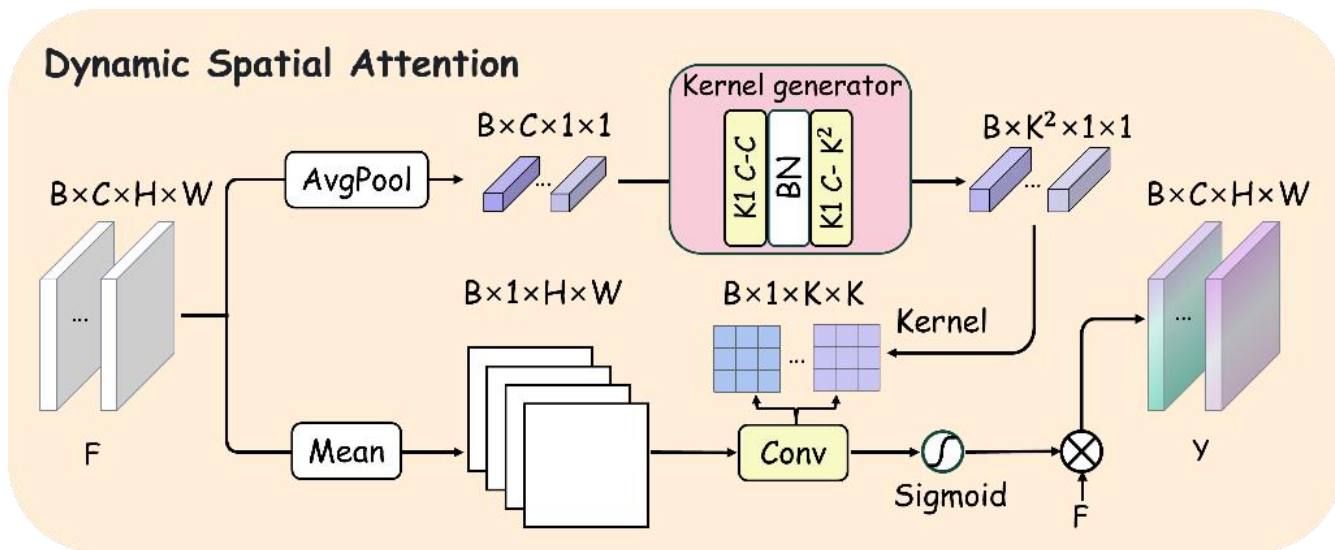
潜在背景编码器模块 (LBEM)

目标动态提取模块 (DTEM) 和参数生成器

图像动态重建模块 (DIRM)

3. DRPCA-Net

DSA 结构图:

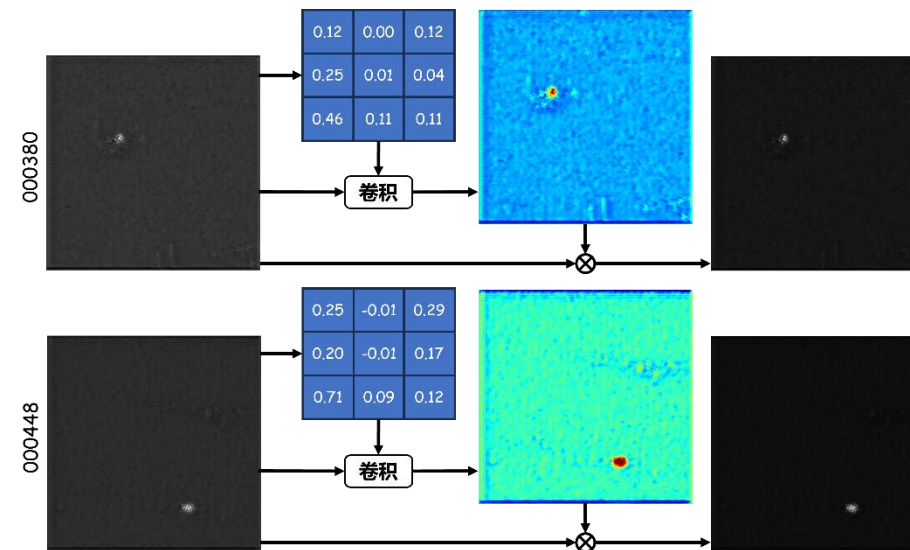


$$1. \quad W_{ds} = \text{Conv}_2 \left(\delta \left(\text{Conv}_1 \left(\text{GAP}(\mathbf{F}) \right) \right) \right)$$

$$2. \quad \mathbf{S}_{\text{att}} = \sigma \left(\text{Conv}_{\text{dynamic}} \left(\mathbf{F}_{\text{mean}}, W_{ds} \right) \right)$$

$$3. \quad \mathbf{Y} = \mathbf{F} \odot \mathbf{S}_{\text{att}}$$

过程可视化:



其根据输入特征 ($\mathbf{F}$) 生成特定于样本的卷积核 (W_{ds}), 以计算自适应空间注意力图 ($\mathbf{S}_{\text{att}}$) 从而实现有针对性的空间特征调制, 以增强重建过程中的目标保留。

4. 实验

对比实验:

Methods	Params	SIRST V1				NUDT-SIRST				SIRST-Aug				IRSTD-1K (Failure Case)			
		IoU↑	F ₁ ↑	P _d ↑	F _a ↓	IoU↑	F ₁ ↑	P _d ↑	F _a ↓	IoU↑	F ₁ ↑	P _d ↑	F _a ↓	IoU↑	F ₁ ↑	P _d ↑	F _a ↓
Traditional Method																	
Tophat [46]	-	26.93	42.44	88.99	182.7	22.25	36.41	91.21	757.3	16.37	28.14	81.15	160.3	6.64	12.46	81.09	142.4
Max-Median [47]	-	32.56	49.12	94.46	33.88	20.49	34.01	84.97	312.7	16.88	28.89	79.36	82.57	20.76	34.38	85.56	212.1
MPCM [48]	-	22.40	36.60	88.07	408.3	9.25	16.94	67.33	317.1	19.76	33.00	90.92	31.42	15.52	26.88	61.85	109.3
HBMLCM [49]	-	11.95	21.36	79.81	413.2	7.60	14.13	60.74	178.4	2.37	46.37	50.61	1461	12.60	22.38	57.73	100.8
NWMTH [46]	-	27.22	42.80	88.99	25.37	14.36	25.11	71.42	199.1	13.31	23.50	83.76	45.72	14.54	25.39	76.97	24.75
FKRW [50]	-	10.43	18.89	78.89	15.26	9.37	17.13	65.29	86.27	4.12	7.93	60.11	119.9	4.94	9.41	58.41	21.10
TLLCM [5]	-	18.73	31.55	81.65	21.11	12.33	21.96	72.63	119.3	2.60	5.08	24.89	18.28	10.24	18.58	64.60	10.40
IPI [18]	-	32.30	48.83	95.41	392.6	31.71	48.16	87.30	109.4	19.00	31.93	75.79	101.9	26.55	41.96	89.34	71.63
PSTNN [51]	-	36.28	53.24	86.23	74.16	21.69	35.64	68.04	222.1	17.75	30.15	58.59	92.05	21.35	35.19	68.04	270.1
NRAM [52]	-	17.16	29.29	71.13	33.48	12.70	22.54	63.49	27.37	8.56	15.77	58.59	46.81	21.05	34.79	78.89	10.47
NOLC [53]	-	20.35	33.82	83.48	13.66	18.60	31.37	70.68	39.69	8.68	15.97	59.00	66.49	10.90	19.65	64.26	18.30
NIPPS [19]	-	28.36	44.18	81.65	20.94	19.11	32.99	69.52	41.46	9.78	17.82	52.54	83.66	21.08	34.82	76.88	22.24
Deep Learning Methods																	
ACM [11]	0.398	69.67	82.12	98.16	26.79	69.22	81.79	94.39	12.85	71.92	83.67	94.63	204.4	54.35	70.26	84.53	11.01
ALCNet [1]	0.378	67.73	80.85	99.08	7.10	69.14	81.76	96.50	30.82	72.94	84.35	98.07	54.20	57.28	72.71	85.56	13.36
ISNet [43]	0.967	67.11	80.31	99.08	57.66	87.53	93.28	96.50	12.20	72.04	83.74	96.14	27.13	63.42	76.77	88.62	28.24
RDIAN [54]	0.216	68.76	81.52	98.16	36.02	85.24	92.03	97.03	14.27	74.88	85.63	98.07	53.59	63.59	77.64	88.31	12.98
ResUNet [55]	0.904	72.31	83.94	97.24	7.63	90.97	95.23	96.08	5.93	72.23	83.87	97.79	41.32	63.08	77.33	92.09	19.05
ISTDU-Net [56]	2.751	73.73	84.88	99.08	8.87	92.99	96.36	98.09	3.98	74.79	85.56	97.11	30.69	65.25	79.04	91.40	19.74
DNANet [13]	4.697	73.03	84.44	99.08	2.46	93.00	96.37	97.98	9.10	71.96	83.68	96.28	42.33	65.52	79.24	93.47	7.82
MTUNet [26]	8.220	73.71	84.89	98.16	1.06	91.60	95.58	97.56	6.48	73.18	84.50	98.34	58.71	63.65	77.74	91.75	14.88
AGPCNet [44]	12.36	68.84	81.54	98.16	17.57	82.52	90.43	96.71	13.42	74.07	85.11	98.07	32.67	60.18	75.14	90.72	13.44
UIUNet [57]	50.54	74.91	85.63	98.16	0.53	92.51	96.08	96.50	1.70	73.45	84.69	97.79	40.90	65.61	79.17	90.03	6.98
MSHNet [58]	4.066	68.35	81.20	98.16	28.02	77.92	87.50	96.82	16.70	72.97	84.33	93.25	305.3	64.56	78.56	89.45	7.21
RPCANet [15]	0.680	68.37	81.21	94.49	24.49	88.24	93.75	96.40	20.96	71.64	83.48	97.93	97.88	61.23	75.95	89.69	22.48
L ² SKNet [59]	0.899	72.23	83.87	98.16	9.23	93.13	96.44	98.09	5.81	72.75	84.22	96.56	54.82	63.73	77.84	92.78	16.85
SCTransNet [27]	11.19	71.15	83.14	97.24	8.87	89.49	94.45	97.67	10.29	72.75	84.20	97.76	52.04	62.04	76.56	73.47	23.68
PConv [60]	4.064	72.62	84.14	99.08	21.46	78.58	88.00	97.56	14.13	73.55	84.76	97.93	74.81	67.45	80.56	93.19	9.71
Ours	1.169	75.52	86.05	99.08	3.73	94.16	96.99	98.41	2.55	76.50	86.69	98.62	30.60	64.14	78.15	92.09	17.92

训练损失:

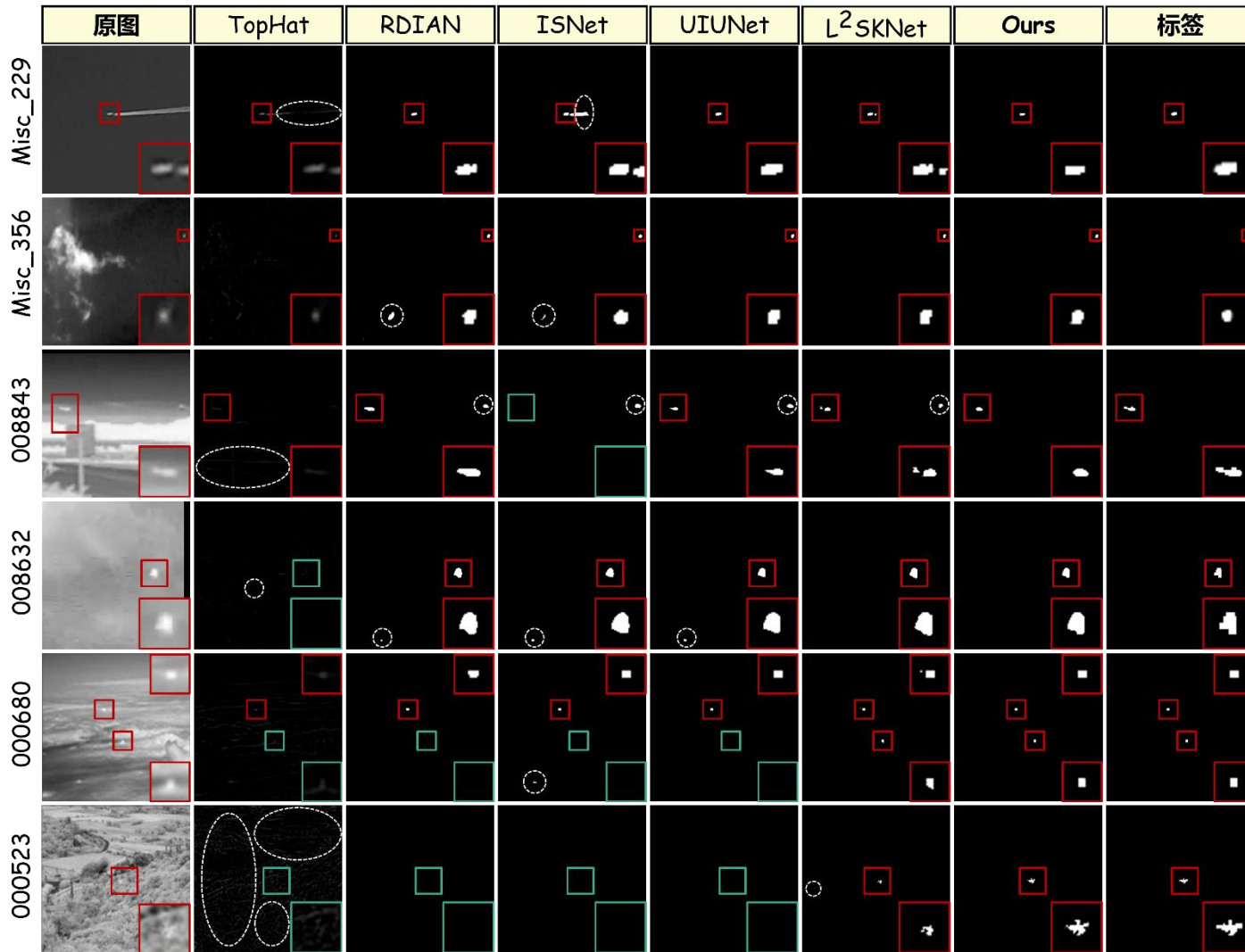
$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda \cdot \mathcal{L}_{\text{rec}}$$

- $\mathcal{L}_{\text{seg}}$: SoftIoU loss
- $\mathcal{L}_{\text{rec}}$: MSE loss
- $\lambda = 0.1$

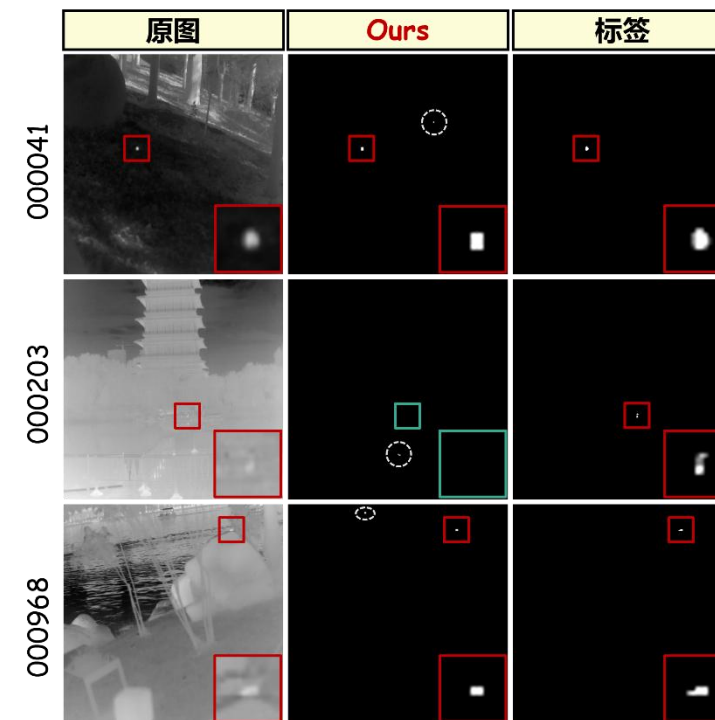
核心组件的消融实验:

Strategy	Module			SIRST V1		NUDT-SIRST	
	ϵ -Gen	γ -Gen	DRG	mIoU↑	F ₁ ↑	mIoU↑	F ₁ ↑
(a)	✗	✗	✗	68.37	81.21	88.24	93.75
(b)	✓	✗	✗	70.36	82.60	90.25	94.87
(c)	✗	✓	✗	70.43	82.65	89.93	94.69
(d)	✗	✗	✓	73.69	84.85	93.47	96.62
(e)	✓	✓	✗	74.74	85.55	91.27	94.86
(f)	✓	✓	✓	75.52	86.05	94.16	96.99

5. 可视化

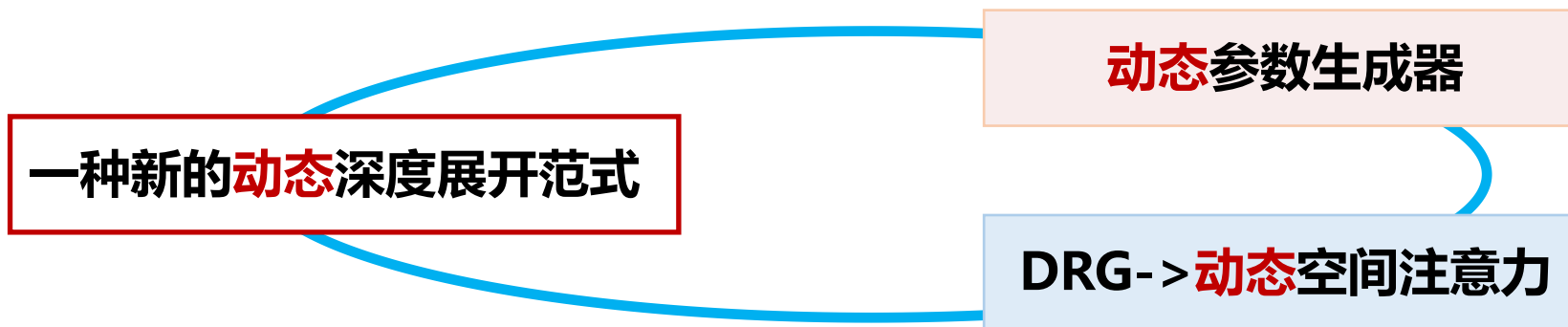


失败案例:



具有挑战性的红外场景的定性比较，正确的检测用红色框标记，漏检用绿色框标记，误报用白色虚线圆圈标记。

6. 总结



谢谢!



Arxiv: <https://arxiv.org/abs/2507.09541>

Code: <https://github.com/GrokCV/DRPCA-Net>